

Course Code	:	MCSL-228
Course Title	:	AI and Machine Learning Lab
Assignment Number	:	MCA_NEW(III)/L-228/Assign/2025
Maximum Marks	:	100
Weightage	:	30%
Last Dates for Submission	:	30th April, 2025 (for January session)
		31st October, 2025 (for July session)

This assignment has 8 Questions for 40 marks. Answer all the questions. Your Lab Record will carry 40 Marks. Rest 20 marks are for viva voce. You may use illustrations and diagrams to enhance the explanation. Please go through the guidelines regarding assignments given in the programme guide for the format of presentation.

Note: You must execute the program and submit the program logic, sample input and output along with the necessary documentation. Assumptions can be made wherever necessary.

- Q1:** Write a Python Program to implement Breadth First Search. **(5 marks)**
- Q2:** Write a Python Program to implement Min-Max Algorithm. **(5 marks)**
- Q3:** Write a Python Program to implement the Backtracking approach to solve N Queen's problem **(5 marks)**
- Q4:** Write a Python Program to implement A* Algorithm. **(5 marks)**
- Q5:** Write a Python Program to implement Naïve Bayes Algorithm for data classification, choose dataset of your own choice. **(5 marks)**
- Q6:** Write a Python Program to implement Polynomial Regression on a dataset of your own choice. **(5 marks)**
- Q7:** Take a Data set as per your choice, implement and execute on different inputs of K-Means clustering algorithm. **(5 marks)**
- Q8:** Write a Python Program to implement FP tree growth Algorithm on a dataset of your own choice. **(5 marks)**